Relations

If we want to describe a relationship between elements of two sets A and B, we can use ordered pairs with their first element taken from A and their second element taken from B. Since this is a relation between two sets, it is called a binary relation.

Definition:

Let A and B be sets. A binary relation from A to B is a subset of $A \times B$ is called relation.

Definition:

Let A and B be two sets. A relation R from A to B is a subset of the Cartesian product $A \times B$.

In other words, a relation R is a set of ordered pairs (a, b) where $a \times A$ and $b \times B$.

If a relation R from A to B, we use the notation aRb to mean the same thing as $(a, b) \in R$. We read it "a is in the relation R to b" or " a is R-related to b".

Note:

If R is a relation from A to B, we write it as:

$$R = \{(a, b): a \in A, b \in B \text{ and } aRb\}$$

- Relations on a Set

If R is a relation from A to itself, we say that R is a relation on A.

Definition:

A relation on the set A is a relation from A to A.

In other words, a relation on the set A is a subset of $A \times A$.

Example 1

Let $A = \{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{(a, b): a \in A, b \in A \text{ and } a < b\}$?

Solution

$$R = \{(1,2), (1,2), (1,4), (2,3), (2,4), (3,4)\}$$

- Domain and Range

Definition: Let R be a relation from A to B, then

The domain of R is the set whose members are the first components (coordinates) of members of R; and is denoted by Dom R.

I.e. Dom $R = \{a: a \in A,\ aRb \text{ for some } b \in B\}$

Or; the domain is the set of all a values in the relation

Definition: Let R be a relation from A to B, then

The Rang of R is the set whose members are the second components (coordinates) of members of R; and is denoted by Ran R.

I.e. Ran $R = \{b: b \in B,\ aRb$ for some $a \in A\}$

Or; the range is the set of all b values in the relation

Note that

$$\text{Dom } R \subseteq A \text{ and Ran } R \subseteq B$$

Example 2 Let $A = \{0, 1, 2, 3\}$, $B = \{2, 4, 6, 8\}$ and $R = \{(a, b): a \in A, b \in A \text{ and } b = 2a\}$ be a relation from A to B.

i. Express R as a set of ordered pairs.
ii. Find the domain and the range of R.

Solution

$R = \{(1,2), (2,4), (3,6)\}$

Dom $R = \{1, 2, 3\}$

Ran $R = \{2, 4, 6\}$

Example 3

Let $R = \{(a, b): a \in A, b \in A \text{ and } b + a = 7\}$ be a relation from $A = \{2, 3, 5\}$ to $B = \{1, 2, 3, \ldots, 10\}$.

i. Write R as a set of ordered pairs.
ii. Find the domain and the range of R.

Solution

$R = \{(2,5), (3,4), (5,2)\}$

Dom $R = \{2, 3, 5\}$

Ran $R = \{5, 4, 2\}$

- The inverse relation

Definition: If $R = \{(a, b): a \in A, b \in B \text{ and } aRb\}$ is a relation from A to B. Then the inverse R^{-1} of R is the relation from B to A and is dented by

$$R^{-1} = \{(b, a): b \in B, a \in A \text{ and } aRb\}$$

Note that

i. R^{-1} can be obtained be reversing the pairs of R
ii. Dom R = Ran R^{-1}, Ran R = Dom R^{-1}

Example 4

Let $R = \{(a, b): a \in A, b \in A \text{ and } a < b\}$ be a relation from $A = \{1, 4, 5\}$ to $B = \{2, 3, 4, 5\}$.

i. Express each of R and R^{-1} as a set of ordered pairs.

ii. Find the domain and the range of both R and R^{-1}.

Solution

$R = \{(1,2), (1,3), (1,4), (1,5), (4,5)\}$

$R^{-1} = \{(2,1), (3,1), (4,1), (5,1), (5,4)\}$

Dom $R = \{1, 4\}$ Ran $R = \{2, 3, 4, 5\}$

Dom $R^{-1} = \{2, 3, 4, 5\}$ Ran $R^{-1} = \{1, 4\}$

- Representation of Relations

There are many ways to represent a relation $R \subseteq A \times B$. Here we give two of them for such a representation.

i. Coordinate diagram (Graph)

When A and B are finite, a relation R from A to B can be represented by a coordinate diagram, where

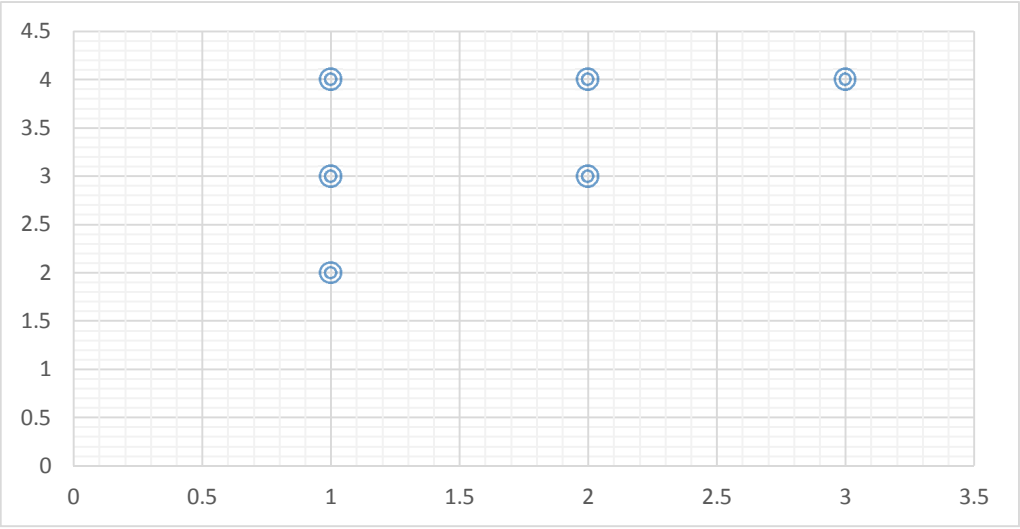
the elements of A are represented on the horizontal axis and the elements of B are represented on the

The set of all points (a, b) which are given in the plane are the elements of R.

Example 5

Draw a coordinate diagram of R in Example 1.

Solution



ii. (Arrows diagram)Mapping

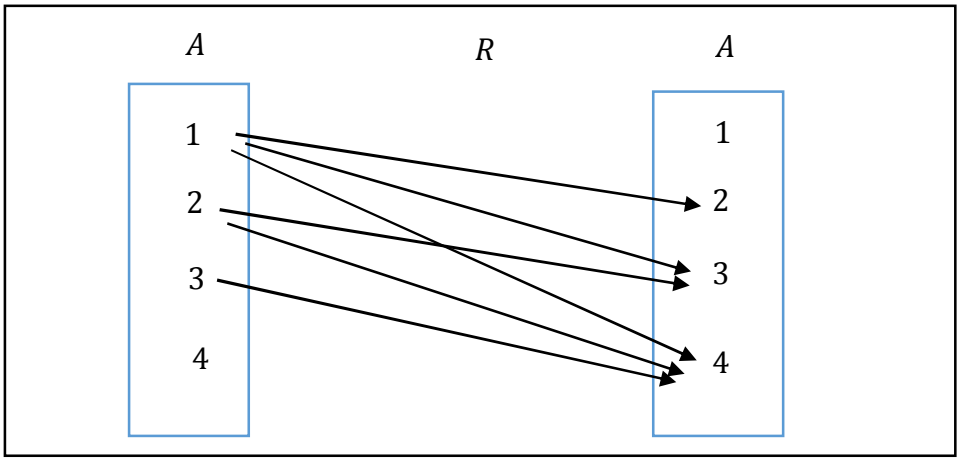
In this method, the elements of A can be put inside a rectangle (or any closed curve) and the elements of B can be put inside another rectangle.

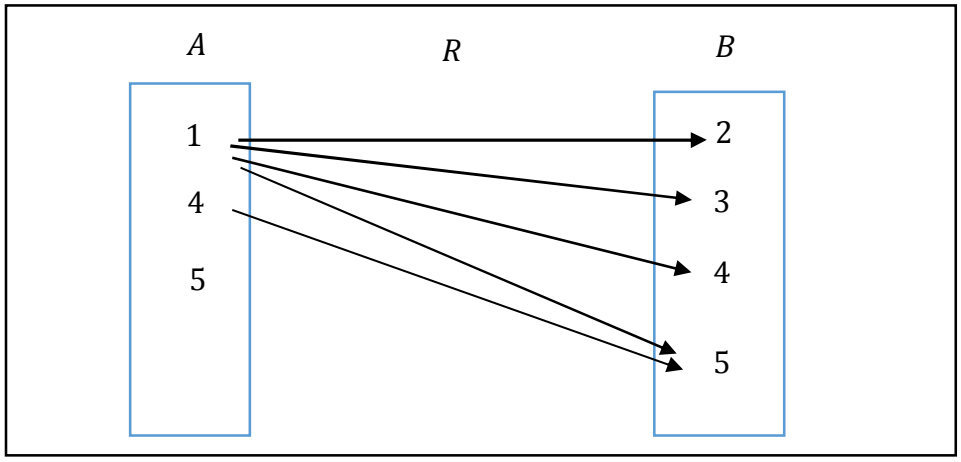
If $(a, b) \in R$, where $a \in R$ and $b \in R$, we draw an arrow from a to b

Example 6

Draw an arrows diagram of R in Examples 1, 2.

Solution





- Composition of Relations

Let R be a relation from A to B and let S be a relation from B to C.

i.e. $R \subset A \times B$ and $S \subset B \times C$.

Then the relation from A to C which consists of all ordered pairs $(a, c) \in A \times C$ such that, for some $b \in B$, $(a, b) \in R$ and $(b, c) \in S$, is called the composition of R and S and is denoted by SoR.

i.e.

$SoR = \{(a, c): a \in A, c \in C, \exists\, b \in B \text{ s.t } (a, b) \in R, (b, c) \in S\}$

Example 7

Let $R = \{(1, a), (1, b), (2, a), (3, d), (4, d)\}$ be a relation from $A = \{1, 2, 3, 4\}$ to $B = \{a, b, c, d\}$ and $S = \{(b, 5), (b, 6), (c, 8), (d, 7)\}$ be a relation from B to $C = \{5, 6, 7, 8\}$. Find SoR.

Solution

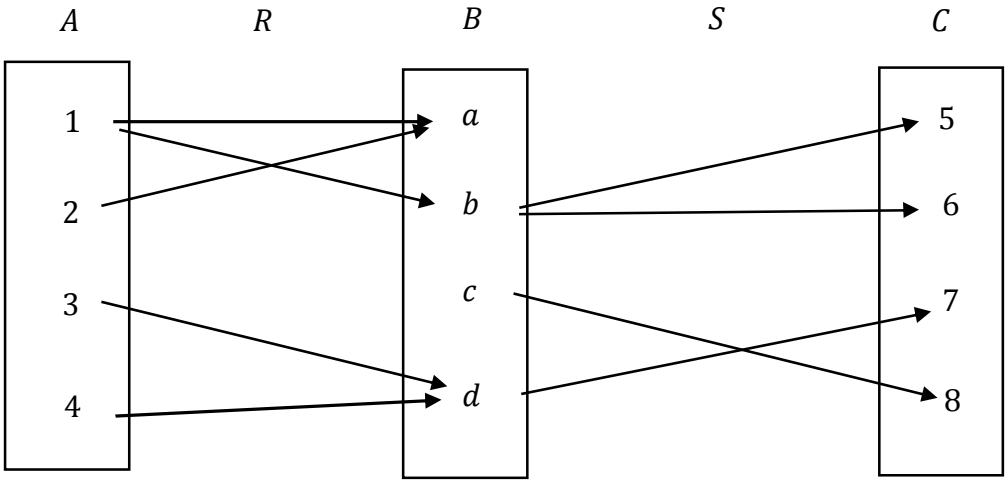
$\because (1, b) \in R$ and $(b, 5) \in S \Rightarrow (1,5) \in SoR$

$\because (1, b) \in R$ and $(b, 6) \in S \Rightarrow (1,6) \in SoR$

$\because (3, d) \in R$ and $(d, 7) \in S \Rightarrow (3,7) \in SoR$

$\because (4,d) \in R$ and $(d,7) \in S \Rightarrow (4,7) \in SoR$

$\therefore\ SoR = \{(1,5),(1,6),(3,7),(4,7)\}$



- Kind of Relations

i. Reflexive relation

Definition: Let R be a relation on a set A.

Then R called reflexive if $(a,a) \in R$ for every $a \in A$.

i.e. R is reflexive if aRa for every $a \in A$.

Example 8 Consider the relation

$R = \{(1,1),(1,2),(2,3),(3,3),(4,4)\}$
on $A = \{1,2,3,4\}$.

Determine whether or not R is reflexive.

Solution

R is not reflexive, since $2 \in A$ but $(2,2) \notin A$.

Example 9

Are the following relations on $\{1,2,3,4\}$ reflexive?

(1) $R = \{(1,1),(1,2),(2,3),(3,3),(4,4)\}$ No.

(2) $R = \{(1,1),(2,2),(3,3),(4,4)\}$ Yes.

(3) $R = \{(1,1),(2,2),(3,3)\}$ No.

ii. Symmetric relation

Definition: Let R be a relation on a set A.

Then R called symmetric if whenever $(a,b) \in R$ then $(b,a) \in R$.

i.e. R is symmetric if $(a,b) \in R \longrightarrow (b,a) \in R$

or if $aRb \longrightarrow bRa$ for all $a,b \in A$.

Example 10

Let $R = \{(1,1),(2,3),(3,2),(3,3)\}$ be a relation on $A = \{1,2,3\}$. Is R symmetric.

Solution

Since $(1,1) \in R \longrightarrow (1,1) \in R$

Since $(2,3) \in R \longrightarrow (3,2) \in R$

Since $(3,2) \in R \longrightarrow (2,3) \in R$

Since $(3,3) \in R \longrightarrow (3,3) \in R$

Then R is symmetric.

Example 11 Are the following relations on $A = \{1, 2,3,4\}$ symmetric or asymmetric?

(1) $R = \{(1,1), (1,2), (2,1), (3,3), (4,4)\}$

Symmetric

(2) $R = \{(1,3), (3,2), (2,1)\}$ Asymmetric

(3) $R = \{(4,4), (3,3), (1,4)\}$ Asymmetric

iii. Transitive relation

Definition: Let R be a relation on a set A.

Then R called transitive if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$ for all $a, b, c \in A$.

i.e. R is transitive if aRb and $bRc \longrightarrow aRc$

Example 12

Let $R = \{(1,1), (2,3), (3,2)\}$ be a relation on $A = \{1, 2,3\}$. Is R transitive.

Solution

Since $(3,2) \in R$ and $(2,3) \in R$ but $(3,3) \notin R$.

Then R is not transitive.

iv. Equivalence relation

Definition: Let R be a relation on a set A.

Then R called an equivalence relation on A if R is reflexive, symmetric, and transitive.

Example 13

Let $R = \{(1,1), (2,2), (3,3)\}$ be a relation on $A = \{1, 2,3\}$. Prove that R equivalence relation on A.

Solution

Since $(1,1) \in R$ ⟹ $(1,1) \in R$

Since $(2,2) \in R$ ⟹ $(2,2) \in R$

Since $(3,3) \in R$ ⟹ $(3,3) \in R$

Then R is reflexive.

Since $(1,1) \in R \Rightarrow (1,1) \in R$

Since $(2,2) \in R \Rightarrow (2,2) \in R$

Since $(3,3) \in R \Rightarrow (3,3) \in R$

Then R is Symmetric.

Since $(1,1) \in R$ and $(1,1) \in R \Rightarrow (1,1) \in R$

Since $(2,2) \in R$ and $(2,2) \in R \Rightarrow (2,2) \in R$

Since $(3,3) \in R$ and $(3,3) \in R \Rightarrow (3,3) \in R$

Then R is transitive.

$\because R$ is reflexive, symmetric, and transitive.

$\therefore R$ is equivalence.

Example 14

Let A be the set of all lines in a plane. Prove that

$R_1 = \{(a, b): a, b \in A, a \text{ is parallel to } b\}$

is an equivalence relation on A, while

$R_2 = \{(a, b): a, b \in A, a \text{ is perpendicular to } b\}$

Is not an equivalence relation on A.

Solution

For R_1:

i) Since $a \parallel a$ for every $a \in A$, then R_1 is reflexive.

ii) If $a \parallel b$, then $b \parallel a$ and so R_1 is symmetric.

iii) If $a \parallel b$ and $b \parallel c$, then $a \parallel c$. Thus R_1 is transitive.

Therefore, R_1 is an equivalence relation on A

For R_2:

i) Since any line a is not perpendicular to itself, then $(a, a) \notin R_2$. Hence R_2 is not reflexive and therefore R_2 is not an equivalence relation on A.

Note that:

R_2 is symmetric because if $a \perp b$, then $b \perp a$.

Also, R_2 is not transitive.

Exercises

1) If $A = \{1, 2, 3\}$, $B = \{1, 3, 4, 6, 9\}$ and $R = \{(a, b): a \in A, b \in A \text{ and } b = 3a\}$ be a relation from A to B.
   i. Express each of R and R^{-1} as a set of ordered pairs.
   ii. Find the domain and the range of both R and R^{-1}.
   iii. Draw a coordinate diagram of R and R^{-1}.
   iv. Draw an arrows diagram of R and R^{-1}.

2) If $R = \{(a, b): a, b \in A, \text{ "}a + b = 6\text{"}\}$ is a relation from $A = \{0, 1, 2, 3, 4, 5, 6, 7\}$.
   i. Write each of R and R^{-1}.
   ii. Find the domain and the range of both R and R^{-1}.
   iii. Show the Kind of R and R^{-1} .
   iv. Find $Ro\ R^{-1}$ and $R^{-1}oR$.

3) If $A = \{0, 1, 2, 3, 4, 5, 6, 7\}$ and $R = \{(a, b): a, b \in A, \text{ "3 divides } a - b\text{"}\}$ be a relation on A.
   i. Write R as a set of ordered pairs.
   ii. Prove that R is an equivalence relation on A.